1. Experimental Section

1.1 Materials

Pyrrolidine-1-sulfonyl fluoride (PSF) was provided by Zhuhai Smoothway Electronic Materials Co., Ltd. 1-((trifluoromethyl)sulfonyl)pyrrolidine (TPSF) solvent were synthesized according to reference 1[1]. Ethanesulfonyl Fluoride (FSE) solvent and N,N-Diethylsulfamoyl Fluoride (FSNDE) were purchased from the Shanghai Titan Technology Co., Ltd. Lithium bis(fluorosulfonyl)imide (LiFSI), lithium hexafluorophosphate ($LiPF_6$), ethylene carbonate (EC), ethyl methyl carbonate (EMC), N,N-dimethylsulfamoyl Fluoride (FSA), 20 μm-thick lithium foil (Φ=15.8 mm), polyvinylidene difluoride (PVDF), and conductive carbon black (Super C45) were purchased from Guangdong Canrd New Energy Technology Co., Ltd. $LiNi_{0.88}Co_xMn_yO_2$ ($x+y$=0.12, NCM88) powder was purchased from BTR New Material Group Co. All the solvents were treated by 4 Å molecular sieves at least 72 hours to remove residual water before use.

1.2 Preparation of electrolyte and electrode

All the electrolyte were prepared and stored in an Ar-filled glovebox ($[O_2]$ < 0.1 ppm, $[H_2O]$ < 0.1 ppm). 0.18706 g LiFSI was dissolved in 1 g PSF solvent to obtain 1 m LiFSI/PSF electrolyte (abbreviated as DVA electrolyte). A carbonate-based electrolyte, 1 M $LiPF_6$ in EC/EMC (3:7 volume:volume), was used as the reference electrolyte. 1 m LiFSI/FSE,1 m LiFSI/FSNDE and 1 m LiFSI/FSA electrolytes were also prepared by dissolving LiFSI salt into the FSE, FSNDE, and FSAsolvent, respectively. To prepare the cathode electrode, a slurry was formulated by mixing NCM88, Super C45, and PVDF in a mass ratio of 94:3:3 using N-methylpyrrolidone as the solvent. The slurry was then cast onto aluminum foil using a doctor blade, followed by drying at 120 ºC for 12 h under vacuum. The dried electrodes were subsequently punched into round disks for coin cell assembly. The mass loading of the active material (NCM88) was approximately 7.5~8.5 mg cm^{-2}.

1.3 Electrochemical measurements

CR2032 coin cells were assembled in the glove box under an argon atmosphere ($[O_2]$ < 0.1 ppm, $[H_2O]$ < 0.1 ppm). For the Li-metal stripping/plating Coulombic efficiency (CE) evaluation, Li||Cu cells were tested at 0.5 mA cm^{-2} and 0.5 mAh cm^{-2} with 30 μL electrolyte. For evaluating the Li-metal CE by Aurbach method, the Li deposits of 5 mAh cm^{-2} were deposited on bare Cu subtracted, followed by fully stripped to 1.0 V at 0.5 mA cm^{-2}. Afterward, 0.5 mAh cm^{-2} Li deposits were cycled for 50 cycles. Li||Li symmetric cells with different electrolytes were tested at 0.5 mA cm^{-2} and 0.5 mAh cm^{-2}. Li||NCM88 full cells were assembled with Li metal foil (20 um-thick) anode, PP separator and NCM88 cathode (~8.3 mg cm^{-2}). The areal capacity and N/P ratio of Li||NCM88 cells are approximately 1.7 mAh cm^{-2} and 2.5, respectively. The electrolyte volume was 30 μL per coin cell. Galvanostatic charge/discharge and rate performance were conducted on the multichannel battery testing system (Neware BTS-53) within 3~4.6 V. Pouch cells (5 cm × 8 cm) were purchased from Atomfair LLC with NCM811 (12 mg cm^{-2}) and pure Li metal (50 μm, without Cu foil) as the cathode and anode, respectively. Pouch cells were charged and discharged at 1C with 4.3 V cutoff voltage, which exhibit an areal capacity of ~2.76 mAh cm^{-2} and N/P ratio of 3.7, respectively. The

electrochemical floating test was performed in coin cells with NCM88 cathode and Li metal anode in different electrolytes. The cells were first charged to 4.6 V at 0.1C and then maintained for 20 h with the current recorded by the LAND system.

1.4 Characterizations

To assess the wettability of electrolyte, contact angle measurement was carried out on a contact angle meter (SDC-200S, SINDIN). Ionic conductivities were examined by Mettler Toledo conductivity meter (SD-30) at room temperature. Basic physicochemical property of pure solvent and electrolytes were evaluated. Melting point was measured by using the Differential Scanning Calorimeter (DSC, AT DSC 25). Viscosity was evaluated using a Brookfield Ametek viscometer at room temperature. Flammability of the solvents and electrolytes was evaluated by igniting experiment. Nuclear magnetic resonance (NMR) spectroscopy was used to analyze the coordination between solvents and salts on Bruker Avance III HD (600 MHz) spectrometer and JEOL JNM-ECZ400S/L1 (400 MHz) at room temperature. Coaxial nuclear magnetic tube was assembled by placing a capillary tube filled with chloroform-d_6 ($CDCl_3$-d_6) into the NMR tubes filled with the different targeted electrolytes. Diffusion coefficients were determined via a diffusion ordered spectroscopy (DOSY) technique based on nuclear magnetic resonance (NMR) spectroscopy using a Bruker Avance III HD (600 MHz) spectrometer (Note S2). The measurements were performed with dstebpgps3s impulse sequence with the parameters of D_{20} (diffusion delay, 0.14s) and P30 (gradient pulse duration, 4100 μs). Based on the same parameters of DOSY technique, the solvation force (χ) of PSF solvent in DVA electrolyte was evaluated using EMC and DME as the reference, toluene as an internal reference. the ^{1}H-DOSY and ^{7}Li-DOSY attenuation curves were obtained for toluene (TOL), FEC, and EMC (or DME) in 1:1 PSF:EMC, 1:6.5:6.5 (molar ratio) LiFSI:PSF:EMC (or DME) electrolyte (Note S2). Raman spectra of different solvents and electrolytes were collected on LabRAM HR evolution spectrometer, excited by a 633 nm laser. Postmortem analysis was performed on the Li metals and NCM88 electrodes after cycling in different electrolytes, which were extracted from the Li||Cu and Li||NCM88 cells. Morphologies were observed with Hitachi SU6600 (15 kV) scanning electron microscopy (SEM). The cathode–electrolyte interphase (CEI) on the cycled NCM88 electrode was examined by high-resolution transmission electron microscope (HRTEM, JEM-2100F). Surface chemistry was detected by Thermo Fisher ESCALB Xi^+ X-ray photoelectron spectroscopy (XPS) with Ar sputtering for various times to obtain the depth profile information. More detailed chemical information of the SEI on Li-metal was investigated by time of flight secondary ion mass spectrometer (TOF-SIMS, ION-TOF M6) equipped with a bismuth primary-ion source. The atomic force microscopy (AFM, MFP-3D) was conducted and evaluated the mechanical property of SEI. To detect gas evolution during the charge process at 0.2 C, *in-situ* DEMS were conducted on an on–line gas analysis mass spectrometer (Hiden HPR-40 EGA). An *in-situ* cell equipped with two poly(ether-ether-ketone) capillary tubes as the gas inlet and outlet was employed to detect the partial pressure changes of the gas products under argon purge. The battery was first flushed with Ar gas at a constant flow rate of 1.5 mL min^{-1} for 3 h to calibrate the baseline.

To observe the microstructural degradation, the ion milling measurement was conducted on Section sample polishing instrument (JEOL, IB-19530CP). The structural evolution was analyzed on *in situ* X-ray diffraction (XRD) with a Bruker DEEP-XRD-R2. The testing employed a scan speed of 6° min^{-1} with a 2θ range of 10–80°. Inductively coupled plasma mass spectrometry (ICP-MS, i CAP Q) measurements was used to detect the transition metal dissolution. Spherical aberration corrected transmission electron microscope (AC-TEM, HITACHI, HF5000) was used to characterize the phase structure of CEI layer.

2. Computational method

2.1 Density functional theory (DFT) simulation

The structures of PSF and EC molecules were optimized with B3LYP[2] method and 6-311+G(d, p) basis set via DFT-D3 (BJ) empirical dispersion correction[3] by Gaussian 16 package.[4] The molecular electrostatic potential (MESP) was plotted in GaussView software. The corresponding values of minimum/maximum points was plotted by VMD 1.9.3[5] and Multiwfn 3.7.[6]

The DFT calculations were performed using the ORCA 6.0.0 software[7]. To fit the restrained electrostatic potential (RESP) charges, the molecular structure was optimized utilizing the M06-2X[8] functional combined with the def2-TZVP[9] basis set. Given that the M06-2X functional primarily accounts for mid-range dispersion effects, the D3zero[10] dispersion correction was applied to enhance the accuracy of the calculations. Subsequently, single-point energy calculations were conducted using the B3LYP[11] functional with the def2-TZVP[9] basis set, incorporating Grimme's-D3 (GD3BJ) dispersion correction[12], which provided adequately high accuracy for the determination of RESP charges. Post-processing of the DFT results was carried out using Multiwfn 3.7 package[6, 13]. The binding energy (E_{bind}) of Li^+ to the salt anion and solvent molecules is defined as: $E_{bind} = E_{Li^+} + E_{mol} - E_{bound}$. Here, the E_{Li^+} represents the energy of Li^+, E_{mol} refers to the energy of the anion (FSI^-) or the solvent molecules (DME, DOL, and PSF), while E_{bound} denotes the energy of the bound system.

2.2 Molecular dynamics (MD) simulation

2.2.1 Simulation Setup

All MD simulations were carried out using the large-scale atomic/molecular massively parallel simulator (LAMMPS)[14]. A cubic simulation box with initial dimensions of 5.0×5.0×5.0 nm^3 was utilized, ensuring that it was sufficiently large to avoid any size effects. Periodic boundary conditions (PBC) were applied in all three spatial directions. The pure solvent simulation system consisted of 520 PSF molecules. As shown in **Fig. S4**, in the electrolyte simulation system, 80 pairs of LiFSI were added, resulting in a molar ratio of PSF to LiFSI of 6.5 and achieving a concentration of 1.0 mol kg^{-1}. Both systems were randomly initialized using the Packmol package[15].

2.2.2 Nonpolarizable Force Field

The parameters for bond, angle, dihedral, and van der Waals (vdW) interactions of FSI^- and PSF were described using the all-atom optimized potentials for liquid simulations (OPLS-AA) force field[16], which has been successfully applied to determine structure-property relationships in

solvents and electrolytes[17]. As shown in **Table S2** and **Figure S5**, the atomic charge and vdW interactions for Li^+ followed the recommendations of Dang[18], while the atomic charges for FSI^- adhered to the guidelines of Canongia Lopes[19]. The atomic charges for the PSF molecule were calculated using RESP analysis, as implemented in the Multiwfn 3.7 package[6, 13]. The interactions among Li^+, FSI^-, and PSF consisted of both electrostatic and vdW components. The long-range Coulombic interactions were computed using the particle-particle-particle-mesh (PPPM) algorithm with a precision of 0.00001[20]. Meanwhile, the vdW interactions were modeled using the 12-6 Lennard-Jones (LJ) potential, truncated at 1.2 nm, with geometric combining rules applied. Furthermore, the partial charges of each atom in LiFSI were scaled by factors of 0.7, 0.8, and 1.0, respectively.

2.2.3 Polarizable Force Field

Specifically, nonpolarizable simulations have been shown to miss essential physics and interactions that require an explicit description of atomic polarization[21]. To address this limitation, the Drude oscillator model[22] explicitly considers the dynamics of electric dipole moments in polarizable atoms. As shown in **Figure S6**, the Drude particles (DPs) were added around non-hydrogen atoms. Each DP possesses a mass of 0.4 g mol^{-1} and connects to a Drude core (DC) via a harmonic spring with a force constant (k_D) of 1000 kcal/mol/Å^2[23]. The atomic polarizability (α) values, provided in **Table S3**, were calculated through Schröder's workflow[24]. The partial charges (q_D) of the DPs were determined using the formula $\alpha = q_D^2 / k_D$.

To mitigate excessive correlations between DCs and neighboring DPs when two Drude oscillators are in close proximity, Thole damping functions were employed to moderate electrostatic interactions over short distances[25]. The Thole damping parameter was set at 2.60 by default, and the cutoff is 1.2 nm. Additionally, the mixing rule for α was implemented as the geometric average of the atom-specific values. We also applied a scaling factor (k_{ij}) to the LJ parameter ε, calculated as:

$$k_{ij} = \left(1 + 0.25 r_{ij}^2 \frac{Q_i^2 \alpha_j + Q_j^2 \alpha_i}{\alpha_i \alpha_j} + 0.11 \frac{\mu_i^2 \alpha_j + \mu_j^2 \alpha_i}{\alpha_i \alpha_j}\right)^{-1} \quad (1)$$

where r_{ij} is the equilibrium center-of-mass distance between Li^+, FSI^-, and PSF. The parameters Q, α, and μ represent the charge, polarizability, and molecular dipole moment of the species, respectively[26].

2.2.4 Simulation Details

All polarizable and nonpolarizable MD simulations were conducted in three stages: (1) Energy minimization at a temperature of 0 K; (2) Equilibration in an NPT ensemble for 10 ns, employing a Nose-Hoover barostat set to a target pressure of 1 atm and Nose-Hoover thermostats set to a target temperature of 298.15 K; (3) Equilibration in the corresponding NVT ensemble for 60 ns, with the final 50 ns used for structural and dynamic analysis. A time step of 1.0 fs was used to ensure the stability of the simulations. The SHAKE algorithm was employed to constrain high-frequency vibrations of C-H bonds[27]. The scaling factors for LJ interactions among intra-molecular atom pairs were adjusted as follows: 1-2 and 1-3 pairs were set to 0.0, while the scaling factor for 1-4 pairs was set to 0.5. For polarizable simulations, special consideration was given to the thermostat in order to maintain a low temperature for DPs and prevent Drude oscillations from affecting the kinetic energy

of the atoms[28]. For this purpose, a dual Nose-Hoover thermostat was implemented to keep the Drude degrees of freedom at 1 K.

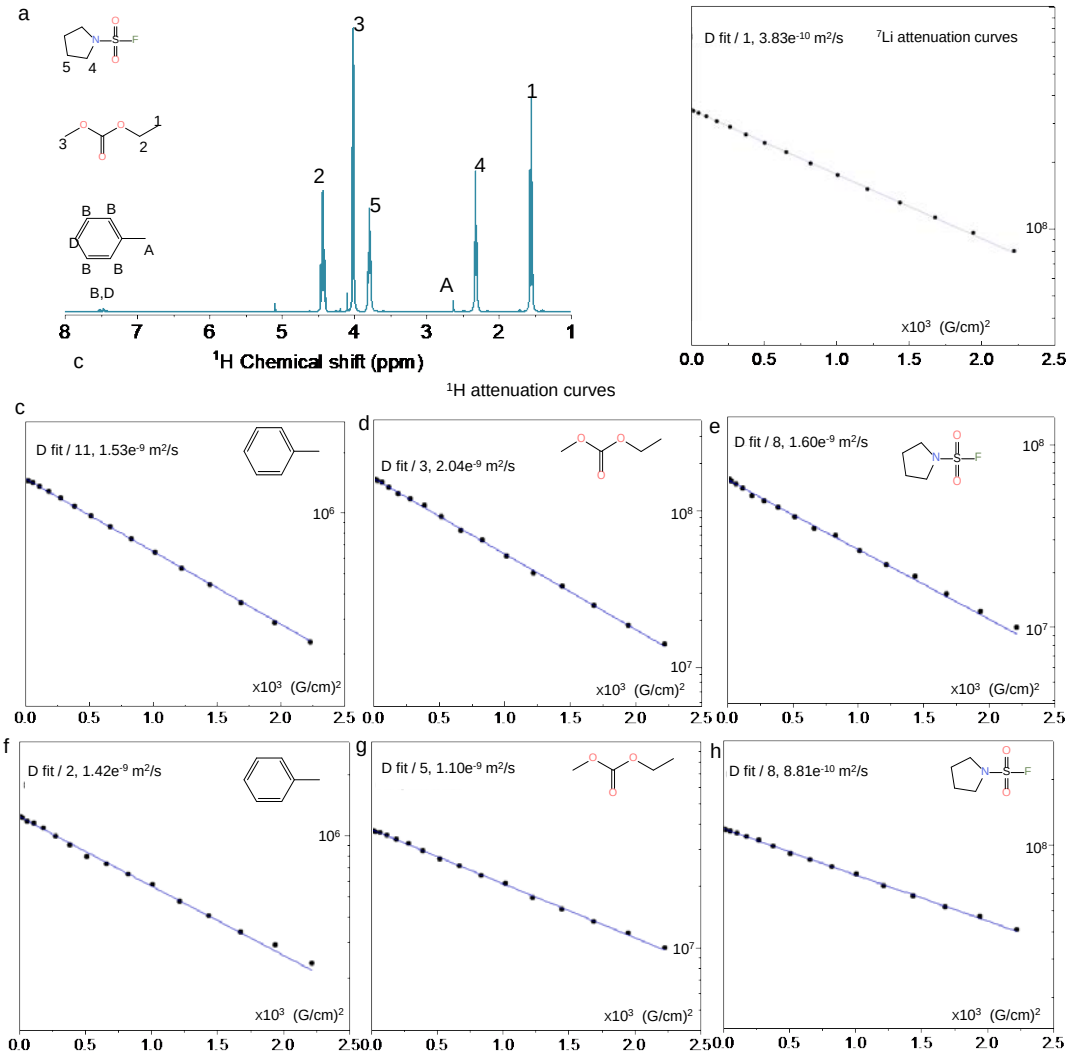


Figure S1 (a) ^{1}H NMR spectrum of LiFSI: PSF: EMC 1: 6.5: 6.5 solution with toluene as internal reference. (b) ^{7}Li DOSY attenuation curves for lithium in 1: 6.5: 6.5 LiFSI: PSF: EMC. (c) ^{1}H DOSY attenuation curves for toluene, EMC and PSF in 1: 1 EMC: PSF. (d) ^{1}H DOSY attenuation curves for toluene, EMC and PSF in 1: 6.5: 6.5 LiFSI: EMC: PSF.

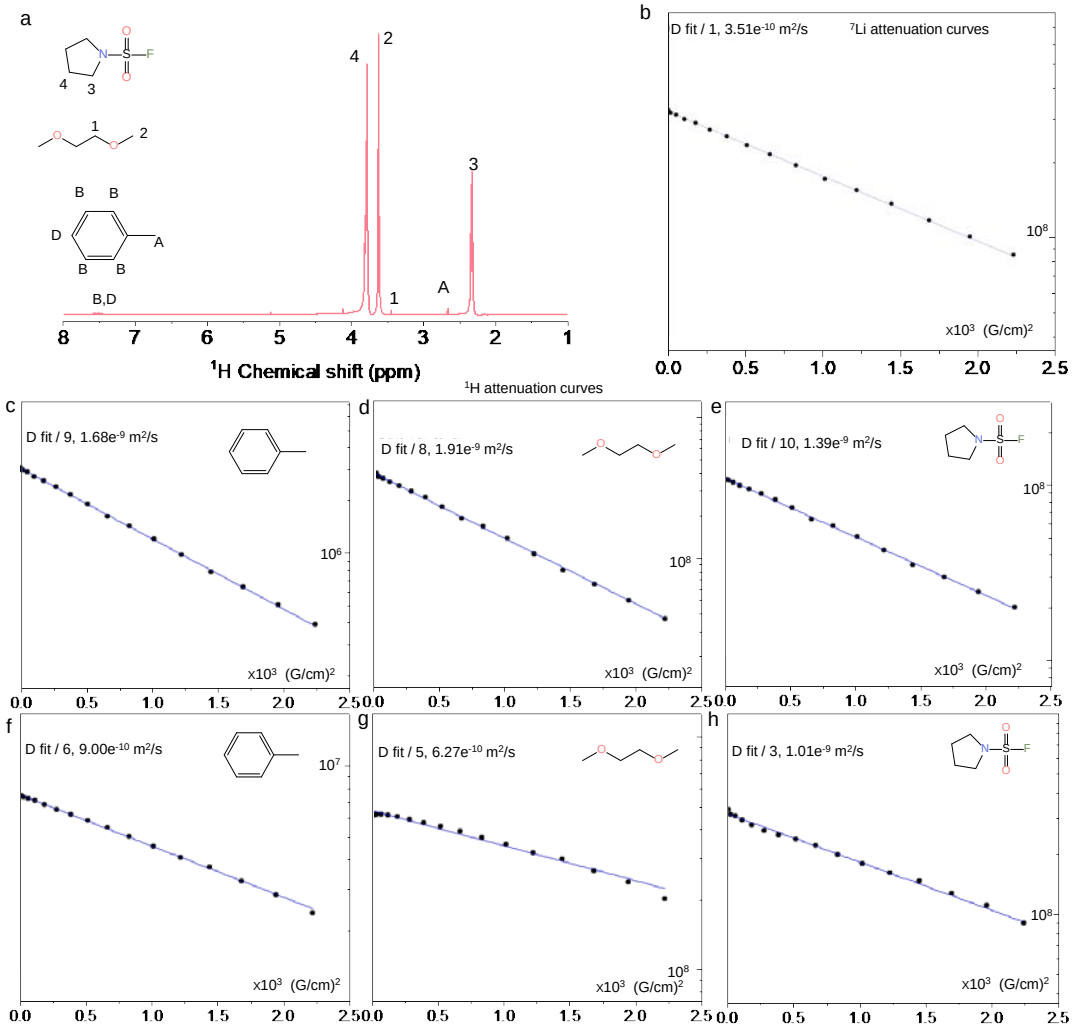


Figure S2 (a) ^{1}H NMR spectrum of LiFSI: PSF: DME 1: 6.5: 6.5 solution with toluene as internal reference. (b) ^{7}Li DOSY attenuation curves for lithium in 1: 6.5: 6.5 LiFSI: PSF: DME. (c) ^{1}H DOSY attenuation curves for toluene, DME and PSF in 1:1 DME: PSF. (d) ^{1}H DOSY attenuation curves for toluene, DME and PSF in 1: 6.5: 6.5 LiFSI: DME: PSF.

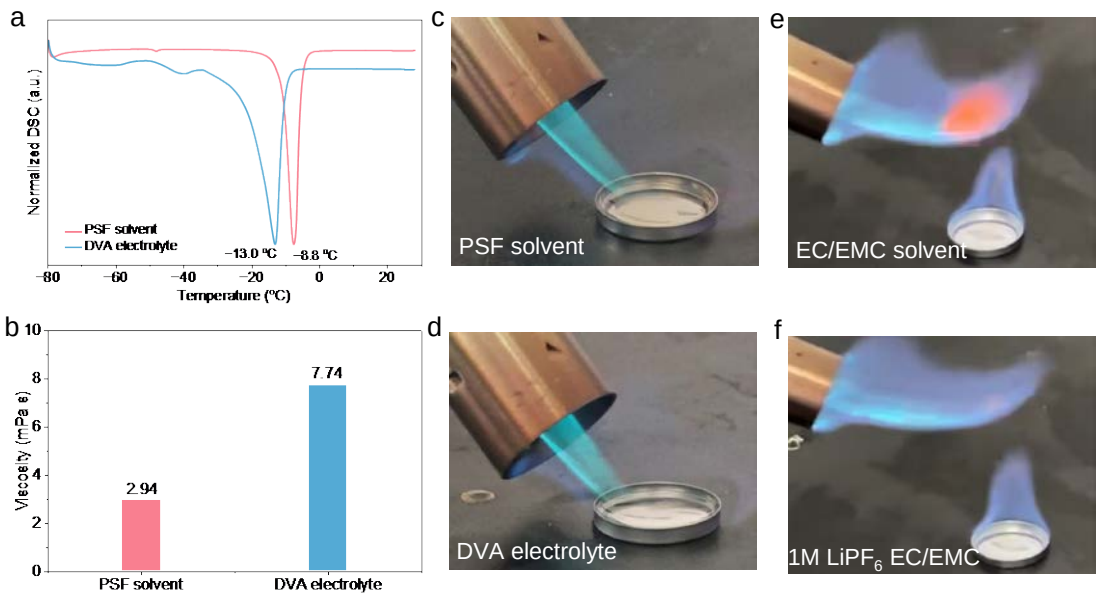


Figure S3 Physicochemical properties of PSF solvent and DVA electrolyte. (a) melting point, (b) viscosity, flammability of (c) PSF solvent, (d) DVA electrolyte, (e) EC/EMC solvent, (f) 1 M $LiPF_6$ EC/EMC electrolyte.

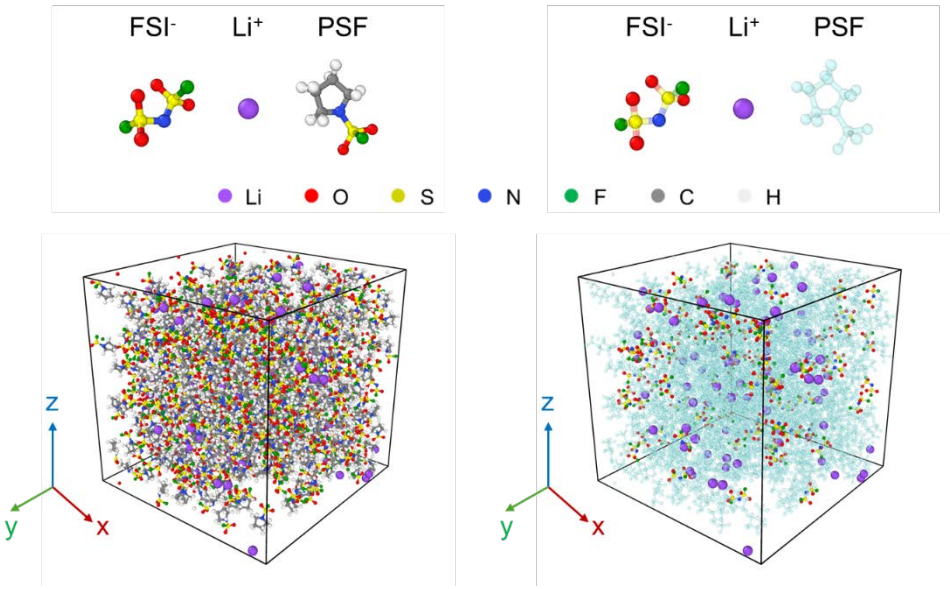


Figure S4. Atomic structures of Li^+, FSI^-, and PSF, along with the initial configuration of the electrolyte simulation system. The colors gray, white, red, green, blue, violet, and cyan represent carbon, hydrogen, oxygen, fluoride, nitrogen, lithium atoms, and the PSF molecule, respectively.

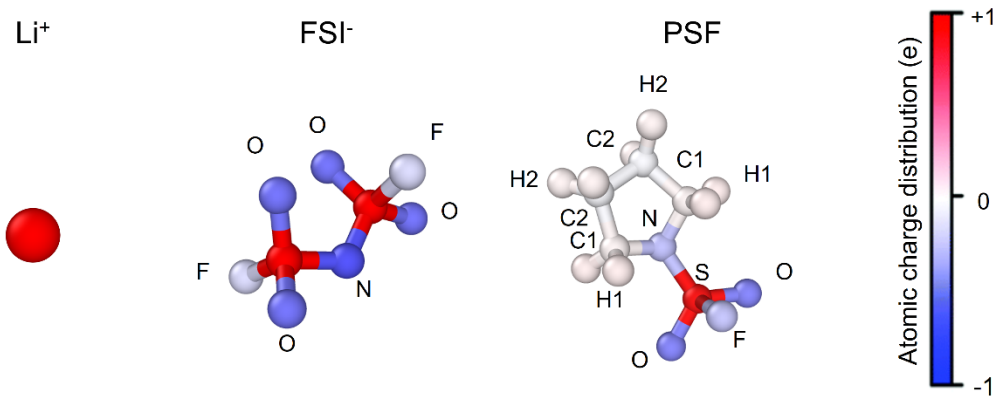


Figure S5. Atomic charge distributions of Li^+, FSI^-, and PSF.

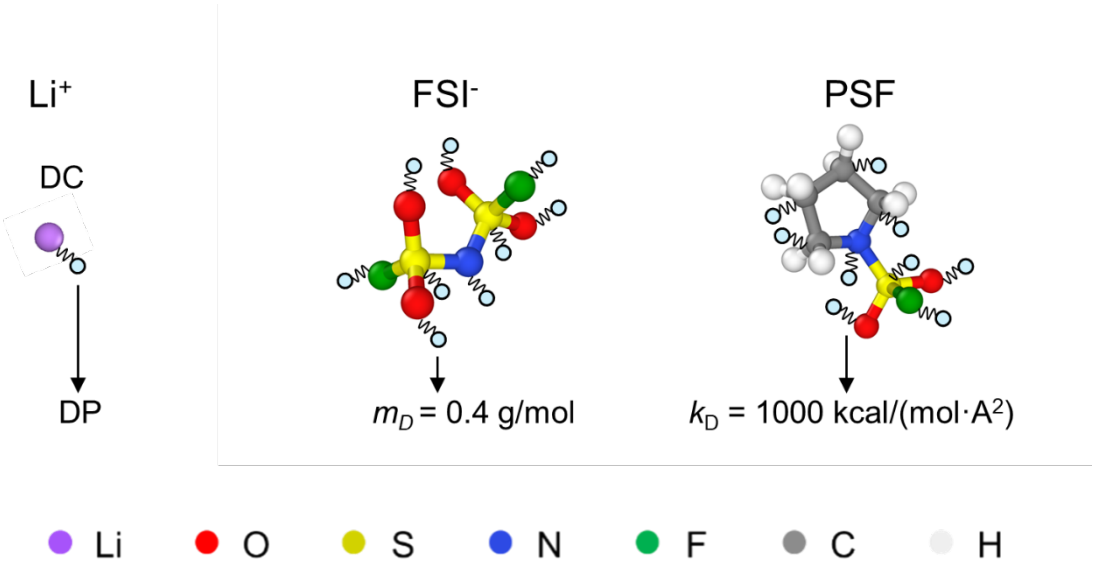


Figure S6. Atomic structures of Li^+, FSI^-, and PSF.

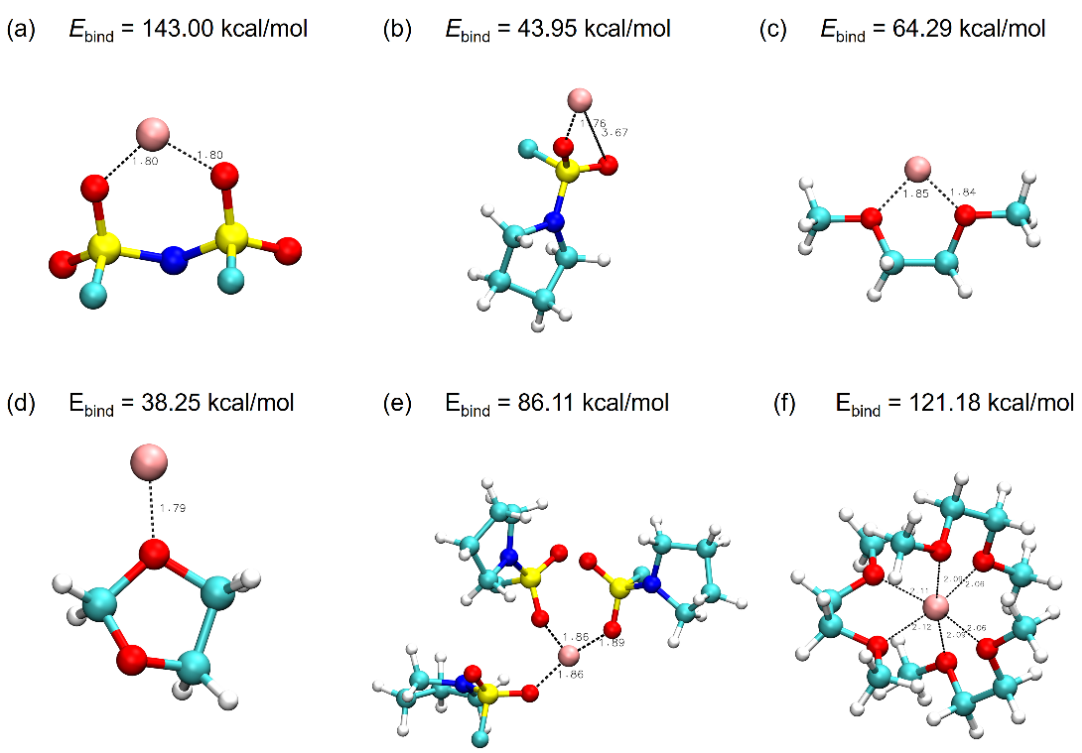


Figure S7. The binding energy (E_{bind}) and DFT-optimized geometry of Li^+ to the salt anion and solvent molecules. (a) Li^+ coordinating to one FSI^-. (b) Li^+ coordinating to one PSF molecule. (c) Li^+ coordinating to one DME molecule. (d) Li^+ coordinating to one DOL molecule. (e) Li^+ coordinating to three PSF molecules. (f) Li^+ coordinating to three DME molecules.

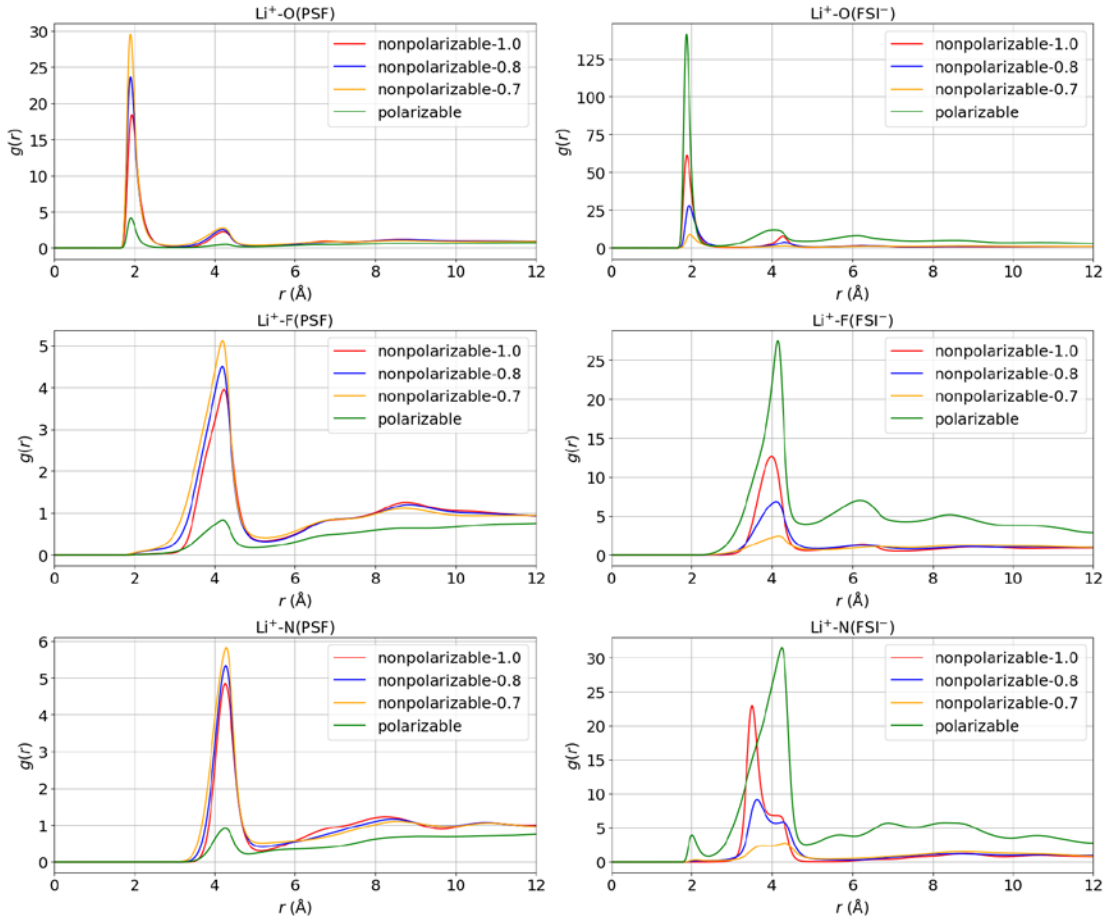


Figure S8. The radial distribution functions (RDFs) between Li^+ and O, F, N atoms in FSI^- and PSF.

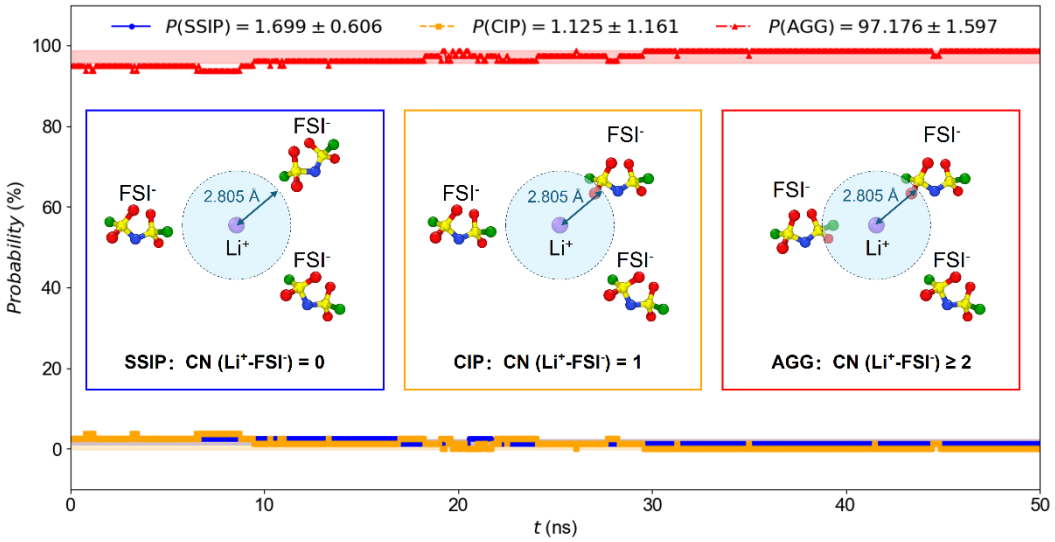


Figure S9. The evolution of different salt–solvent clusters with the simulation time (t). P_{SSIP}, P_{CIP}, and P_{AGG} represent the probabilities of SSIP, CIP, and AGG formation in the simulation system, respectively. Insets clarify the definitions of the different cluster types, with the distance of 2.805 Å determined from the first valley of the RDF for Li^+–O.

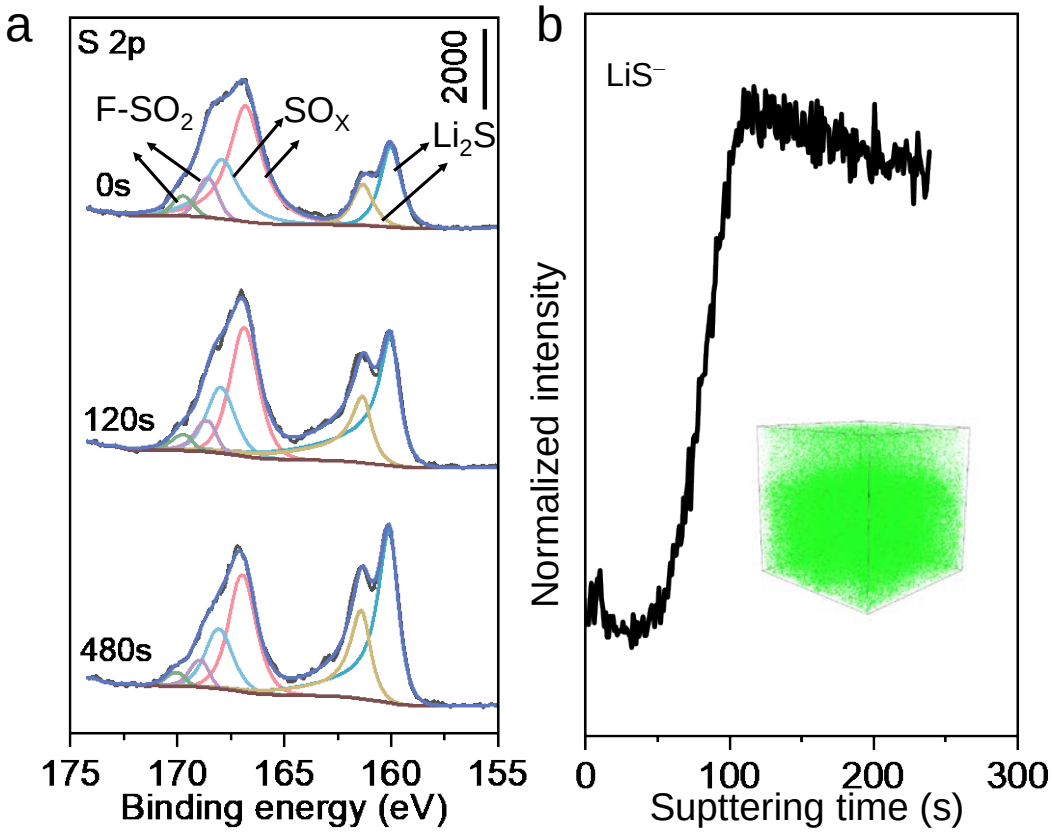


Figure S10 The XPS depth analysis of S 2p (a) and TOF-SIMS analysis of LiS^-, characterized on the Li deposition extracted from the Li||Cu cells with the DVA electrolyte.

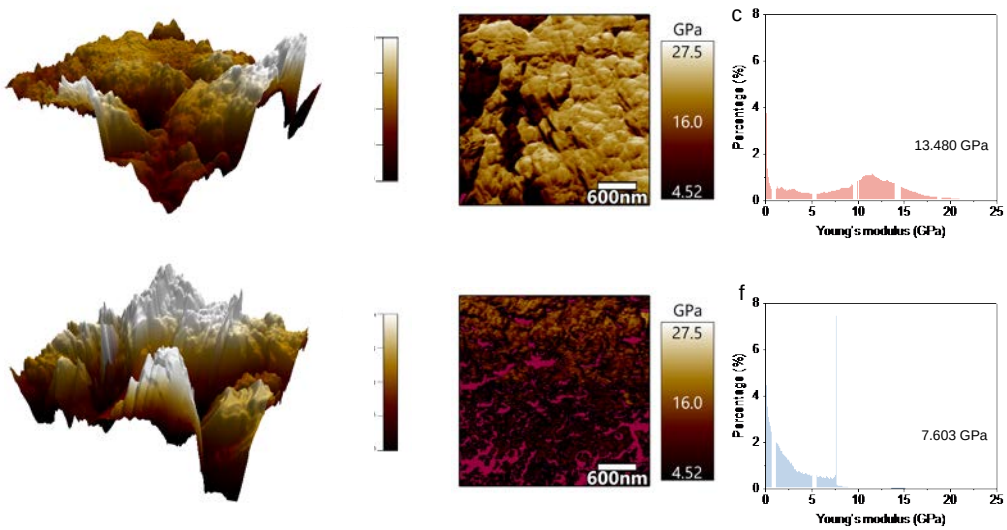


Figure S11 The mechanical properties of SEIs formed in DVA electrolyte (a–c) and carbonated-based electrolyte (d–f).

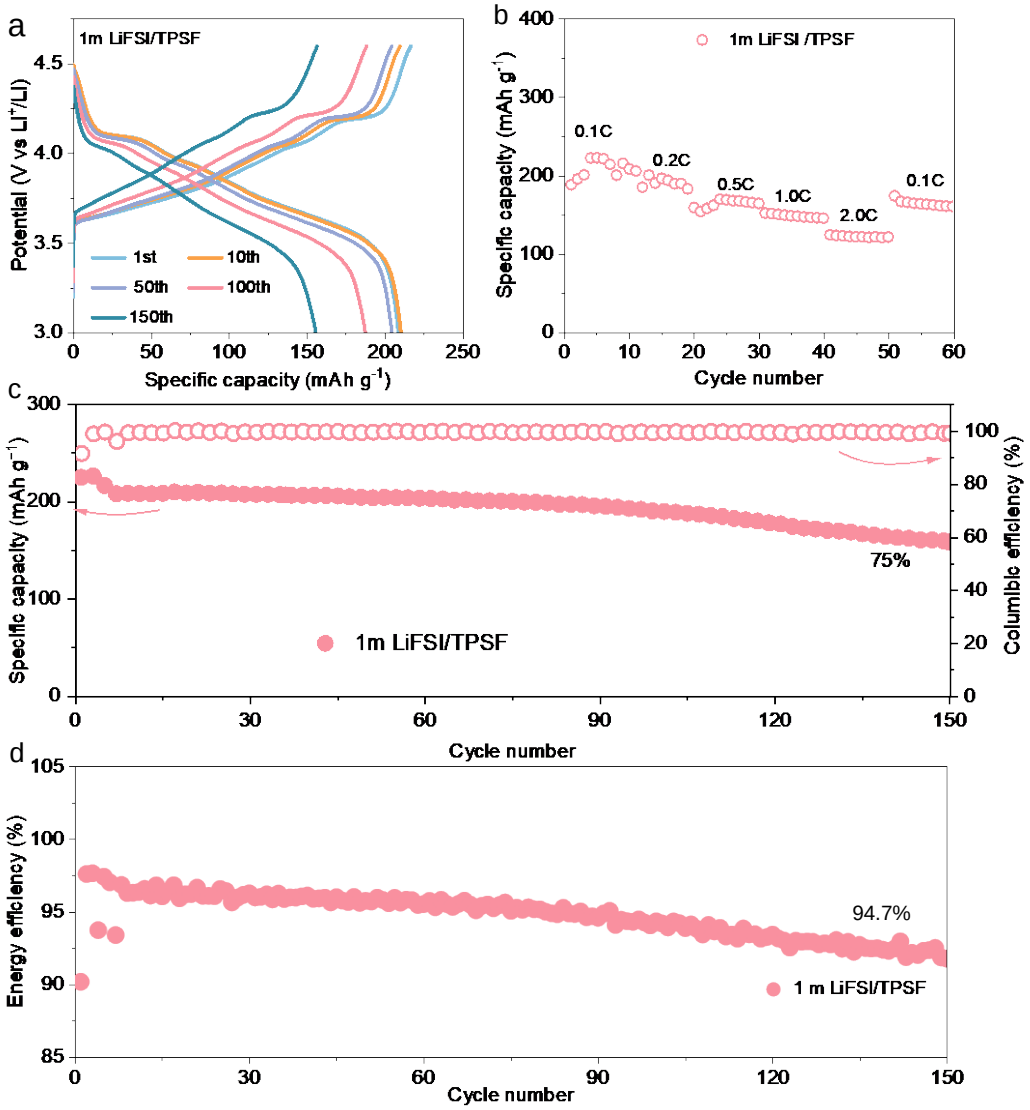


Figure S12 The electrochemical performance of Li||NCM88 cell in 1 m LiFSI/TPSF electrolyte. (a) Voltage profiles at 0.5 C, (b) rate performance, (c) long-term cycling performance and (d) energy efficiency at 0.5 C

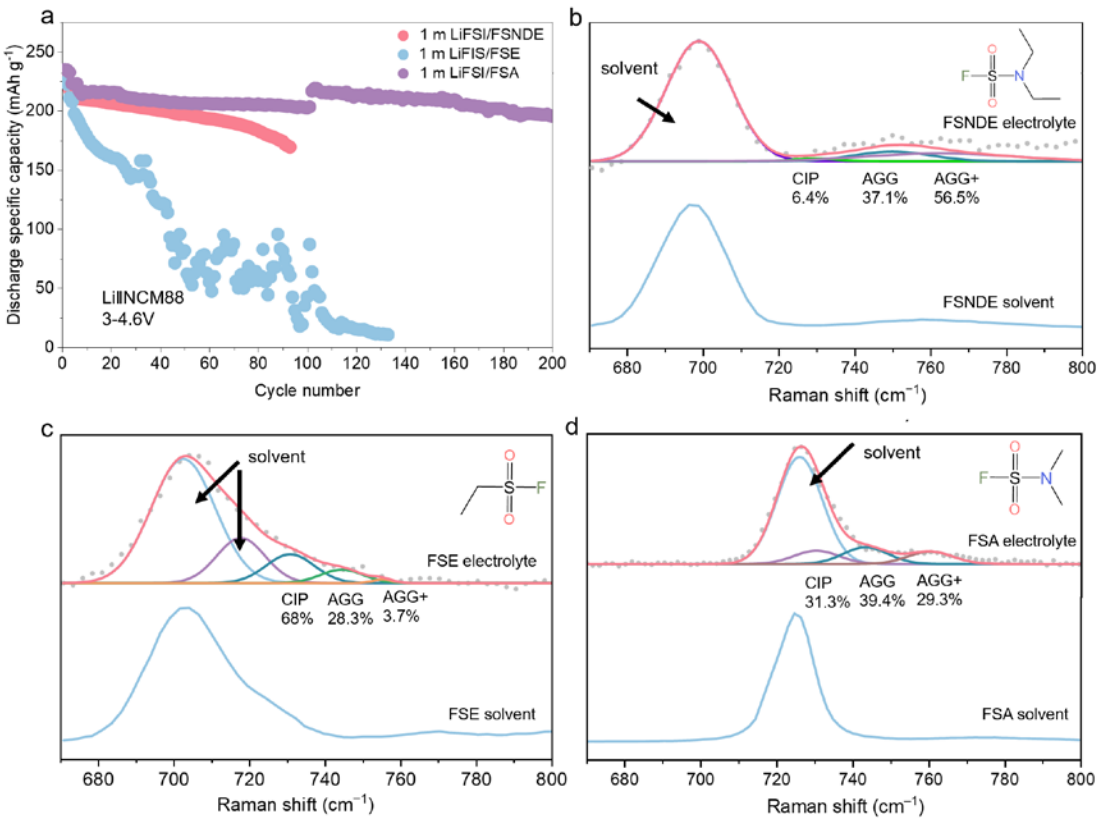


Figure S13. Electrochemical performance Raman spectra. (a) Cycling performance of the Li||NCM88 cells with 1 m LiFSI in FSNDE/FSE/FSA electrolytes at an upper cut-off voltage of 4.6 V. Raman spectra of the pure solvents and electrolytes (b, FSNDE; c, FSE; d, FSA) at 25 ^{o}C. Deconvolution of Li^{+}–FSI^{-} Raman peaks for these electrolytes with peak fits for different cluster types denoted by arrows.

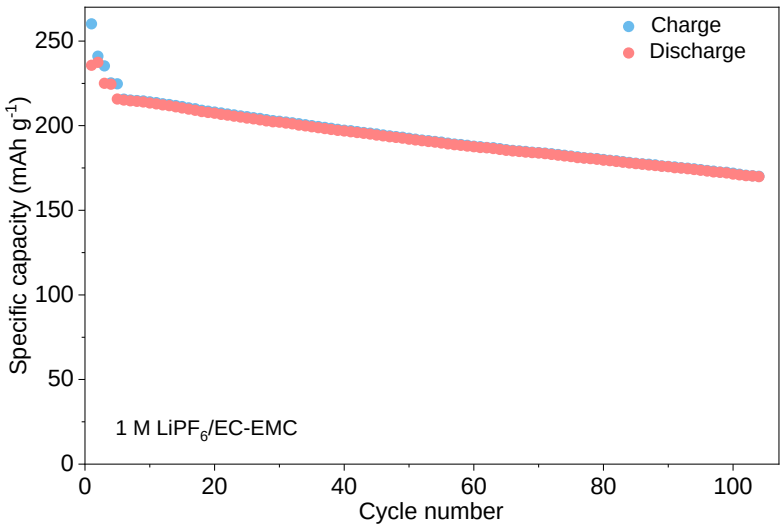


Figure S14 The electrochemical performance of Li||NCM88 cell with ~2 mg cm^{-2} loading in 1 M $LiPF_6$/EC-EMC electrolyte.

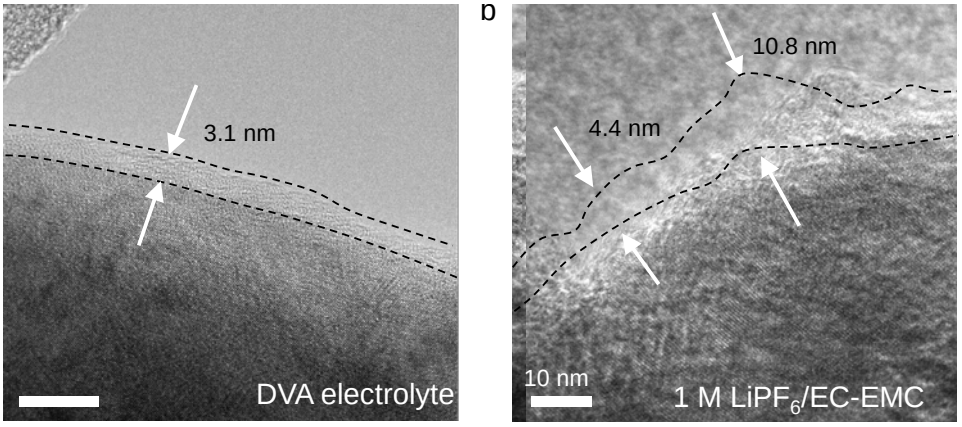


Figure S15 HRTEM images of CEIs after cycling in the DVA (**a**) and 1 M $LiPF_6$/EC-EMC (**b**) electrolytes.

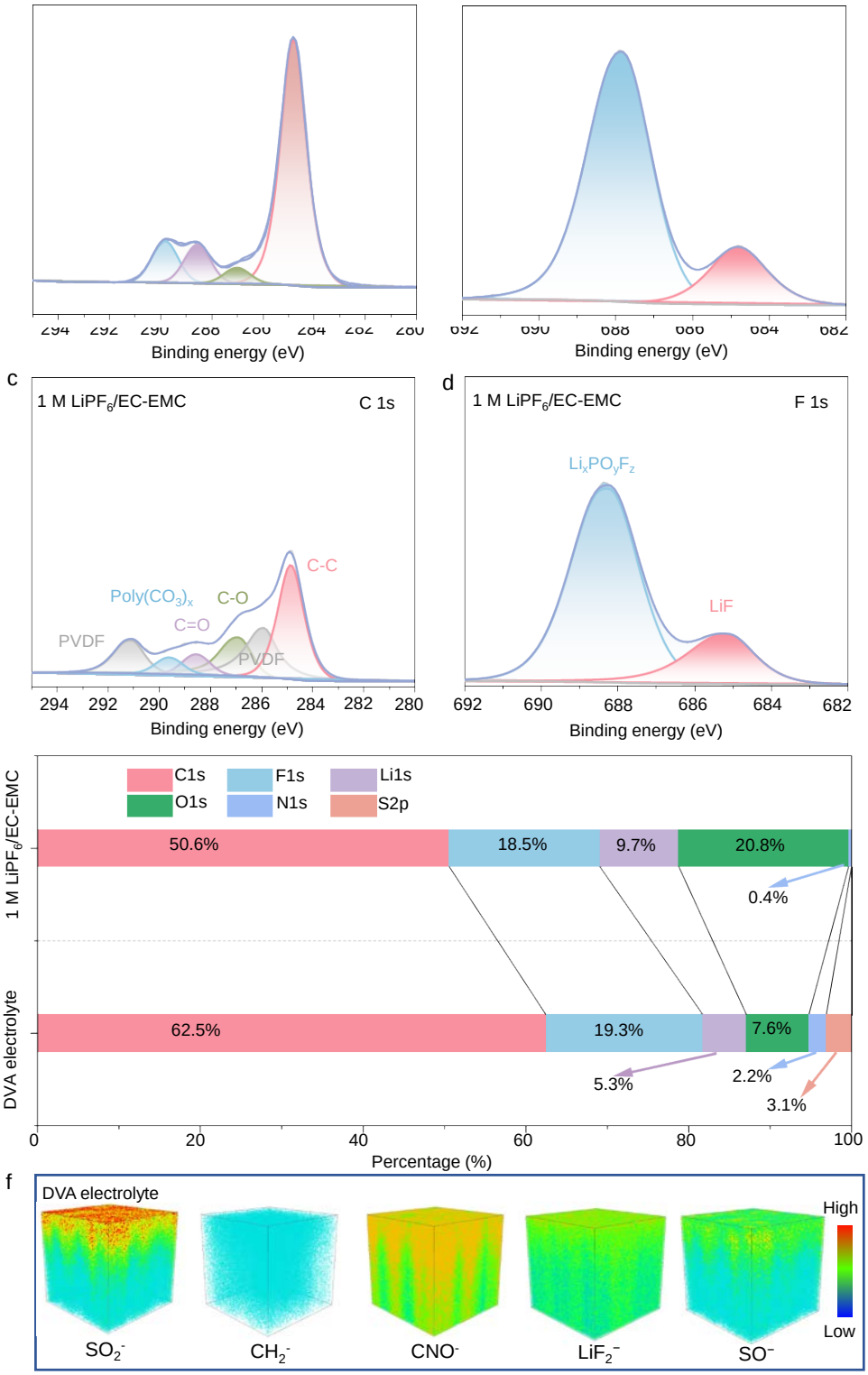


Figure S16 C1s and F1s XPS spectra of CEIs formed in the DVA (a, b) and 1 M $LiPF_6$/EC-EMC (c, d) electrolytes. (e) Atomic ratios of different elements in CEIs by XPS (f) TOF-SIMS 3D reconstructed from the CEI formed in the DVA electrolyte.

Table S1. Coordination ratio α of 1: 6.5: 6.5 LiFSI: PSF: DME electrolyte (I) and 1:6.5:6.5 LiFSI: PSF: EMC electrolyte (II)

	D_{TOL}	D_{DME}	D_{EMC}	D_{PSF}	D_{Li+}	α_{DME}	α_{PSF}	χ
Solvent (I)	1.68	1.91		1.39				$\frac{\alpha_{PSF}}{\alpha_{DME}} = 0.63$
Electrolyte(I)	0.90	0.627		1.01	0.351	1.06	0.67	
Solvent (II)	1.53		2.04	1.60				$\frac{\alpha_{PSF}}{\alpha_{EMC}} = 1.06$
Electrolyte(II)	1.42		1.10	0.881	0.383	0.52	0.55	

Note: α represents the coordination percentage of solvent. D (m^2 S^{-1}) is diffusion coefficient.

Table S2. Atomic charges and Lennard-Jones (LJ) potential parameters for Li^+, FSI^-, and PSF.

Species	Atom Type	Atomic Charge (e)	σ (Å)	ε (kcal mol^{-1})
Li^+	Li	1.0000	1.506	0.165
FSI^-	F	−0.1300	2.950	0.053
	S	1.0200	3.550	0.250
	N	−0.6600	3.250	0.170
	O	−0.5300	2.960	0.210
PSF	H1	0.0641	2.500	0.030
	C1	0.0335	3.500	0.066
	N	−0.2283	3.250	0.170
	C2	−0.0182	3.500	0.066
	S	0.8988	3.550	0.250
	H2	0.0417	2.500	0.030
	O	−0.4563	2.960	0.170
	F	−0.2117	2.900	0.060

Table S3. The atomic polarizability (α) and molecular dipole moment (μ) for Li^+, FSI^-, and PSF.

Species	Atom Type	α (Å^3)	α (Å^3)	μ (Debye)
Li^+	Li	0.019	0.019	0
FSI^-	F	0.471	9.609	1.733
	S	2.442		
	N	0.946		
	O	0.709		
PSF	H1	0.134	11.741	5.456
	C1	1.557		
	N	0.794		
	C2	1.499		
	S	2.107		
	H2	0.134		
	O	0.618		
	F	0.418		

Table S4. The density (ρ) values of pure solvent and electrolyte

System Type	ρ (g cm^{-3})	method	
pure solvent	1.372	MD simulation	nonpolarizable-1.0
	1.247		polarizable
	1.336	Experiment	
electrolyte	1.486	MD simulation	nonpolarizable-1.0
	1.480		nonpolarizable-0.8
	1.475		nonpolarizable-0.7
	1.342		polarizable
	1.392	Experiment	

Table S5. The diffusion coefficient (*D*) values of pure solvent and electrolyte.

System Type	Species	D (10^{-10} m^2 s^{-1})	method	
pure solvent	PSF	0.2616	MD simulation	nonpolarizable-1.0
		5.5580		polarizable
		4.31	Experiment	
electrolyte	Li^+	0.0002	MD simulation	nonpolarizable-1.0
	FSI^-	0.0004		
	PSF	0.0009		
	Li^+	0.0014		nonpolarizable-0.8
	FSI^-	0.0017		
	PSF	0.0024		
	Li^+	0.0040		nonpolarizable-0.7
	FSI^-	0.0050		
	PSF	0.0042		
	Li^+	0.1370		polarizable
	FSI^-	0.1300		
	PSF	3.4000		
	Li^+	1.04	Experiment	
	FSI^-	0.901		
	PSF	1.77		

Note S1: Solvation force (χ)

As a representative example, the electrolyte of $LiPF_6$: EC: EMC 1: 6.5: 6.5 is used to explain the principle of solvation force (χ) calcultion.
According to the Stokes-Einstein equation, the diffusion coefficient of PSF (D_{1PSF}) in PSF: EMC 1:1 solution can be calculated with the followed formula:

$$D_{1PSF}=kT_1/6\pi\eta_1 r_1 \quad (2)$$

k: Boltzmann constant, T_1: temperature, η_1: viscosity, r_1: hydrodynamic radius

After the LiFSI is dissolved in an PSF: EMC 1:1 solution, the Li^+ is solvated by the solvent molecules. The hydrodynamic radius of PSF and the viscosity of the solution will increase. In order to obtain the viscosity of LiFSI: PSF: EMC solution, we added the toluene (as an internal reference) to the PSF: EMC and LiFSI: PSF: EMC solution. Due to the toluene is not participate in coordination with Li^+, the diffusion coefficient of toluene before (D_{1Tol}) and after (D_{2TOL}) the addition of LiFSI is calculated:

$$D_{1TOL}= kT_1/6\pi\eta_1 r_1 \quad (3)$$

$$D_{2TOL}=kT_2/6\pi\eta_2 r_2 \quad (4)$$

Because toluene solvent has extremely low polarity and does not undergo solvation with Li^+, the r of toluene remains unchanged, named the $r_1= r_2$. Therefore, the T and η of PSF: EMC and LiFSI: PSF: EMC soulution are determined by the ratio of D_{1TOL} and D_{2TOL}.

$$D_{1TOL} / D_{2TOL} =T_1\eta_2/T_2\eta_1 \quad (5)$$

For the LiFSI: PSF: EMC solution, the PSF is presented with the formation of Li^+-PSF complexes and free PSF solvent. The diffusion coefficient of PSF(D_{2PSF}) is obtained by the $D_{Co\text{-}PSF}$ and $D_{free\text{-}PSF}$.

$$D_{Co\text{-}PSF} = D_{Li+} \quad (6)$$

$$D_{free-PSF} = \frac{\frac{kT_1}{6\pi\eta_1 r_1}}{\frac{kT_2}{6\pi\eta_2 r_1}} D_{1PSF} = \frac{T_1\eta_2}{T_2\eta_1} D_{1PSF} \quad (7)$$

$$D_{2PSF}=D_{Co\text{-}PSF}\,\alpha+D_{free\text{-}PSF}\,(1-\alpha) \quad (8)$$

Relative solvating power χ is defined as the ratio between the coordination percentage of a test solvent PSF (α) and the coordination percentage of a reference solvent EMC (α_0) as described in equation 8.

$$\chi = \frac{\alpha}{\alpha_0} \quad (9)$$

The diffusion coefficient of PSF: EMC and LiFSI: PSF: EMC are obtained from the DOSY-NMR tests.

Note S2: diffusion coefficient

The diffusion coefficient is fitted by the formula (10), which described the relationship between the signal attenuation and the diffusion coefficient in the pulsed gradient field nuclear magnetic resonance (PFG-NMR) experiment.

$$I = I_0 * e^{\left(-D*(2\pi*\gamma*Gi*LD)^2*\frac{BD-LD}{3}*1e^4\right)} \quad (10)$$

I: The NMR signal intensity under applied diffusion gradients

I_0: the NMR signal intensity without diffusion gradients
D: the diffusion coefficient (m^2/s)
γ: Gyromagnetic ratio (rad/ (s T)). For different atomic nuclei, γ is a constant.
Gi: Gradient amplitude (T/m)
LD: Little delta (δ, s)
BD: Big delta (Δ, s)

During the process of calculating the diffusion coefficient, the signal intensity I_0 without the gradient field and the time parameters *Δ* and *δ* are fixed. Then the gradient field intensity Gi is changed and the corresponding signal intensity *I* are recorded.

The formula (10) is simplified to the following form:

$$ln\frac{I}{I_0} = -D * k \quad (11)$$

$$k = (2\pi)^2\gamma^2 Gi^2\delta^2(\frac{\Delta-\delta}{3}) \quad (12)$$

Plot a scatter graph with ln(I/I₀) as the dependent variable (on the y-axis) and Gi^2 as the independent variable (on the x-axis). The slope value (a) is obtained through linear fitting by the formula (13)

$$a = -D * (2\pi)^2\gamma^2\delta^2(\frac{\Delta-\delta}{3}) \quad (13)$$

The value of the diffusion coefficient (*D*) can be determined based on the known parameters.

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